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Dear Cascade Data Labs,

The nature of scientific inference and its relation to data collection, data analysis, prediction, causality, and decision making continue to be at the center of my professional interests.[1,2] Scientific inference is simply any statement (summary, hypothesis, law, proposition, conclusion, projection, or prediction) that is made beyond the empirical data. Because I am a scientist of an ilk towards the fundamental analysis of systems, structures, and patterns, I am constantly considering how various tools for inference may be applied to better understand the behavior of complex systems. This leads, mostly, to many questions:

- Under what constraints are the statistics and hypothesis testing algorithms of the Neyman-Pearson (Frequentist) school of statistical inference valid for the study of complex systems?
- Can the engines of Bayesian statistics, driven by probability theory, truly provide a typology of inductive inference using the weaker syllogisms?[3] (Given that if A is true then B is true, the weaker syllogisms would be (1) B is true therefore A is more plausible, or (2) A is false therefore B is less plausible.)
- Given that complex systems must be simplified for study and that such simplification must reduce the discriminating power of observation (i.e., the data), is it possible to estimate the error of omission on the simplification?[4]
- Would observation-based, phenomenological investigations into complex adaptive systems benefit from the use of generative models? That is, by starting with a joint probability distribution and then building either a lattice of suspected relationships between observables (variables) as a directed acyclic graph[5] or as a hidden Markov model?[6]

[1] Harold Jeffreys. *Scientific inference*. eng. 1. paperback ed. Cambridge: Cambridge Univ. Press, 2010.
[2] Richard T. Cox. *The algebra of probable inference*. eng. Baltimore [Md]: Johns Hopkins Press, 1961.
[3] E. T. Jaynes and G. Larry Bretthorst. *Probability Theory: The Logic of Science*. Cambridge, UK ; New York, NY: Cambridge University Press, 2003.
[4] William Ross Ashby. *An introduction to cybernetics*. Taylor & Francis, 1955.
[5] David Lunn. *The BUGS book: a practical introduction to Bayesian analysis*. Texts in statistical science. Boca Raton, FL: CRC Press, Taylor & Francis Group, 2013.
[6] Gernot A. Fink. *Markov models for pattern recognition: from theory to applications*. eng. Second edition. Advances in computer vision and pattern recognition. London: Springer, 2014.

- What role will cellular autonoma and agent-based models[7] play in modern data science if Wolfram's Principle of Computational Equivalence and "computational irreducibility" are accepted as a valid?[8]
- Can information theoretic measures and maximum entropy models be useful in predicting behavioral states of complex adaptive systems?[9]
- And verbatim from Pearl, "What empirical evidence is required for legitimate inference of cause-effect relationships?"[10] Is policy analysis truly an exercise in counterfactual reasoning?

There are more.

I have spent the last few years as a doctoral student and IGERT fellow thinking about how to model agroecological mutualisms; the key, I believe, to sustainable agriculture. Although I have passed comprehensive exams and completed all course work in the Earth, Environment, and Society program at Portland State University, I have decided for the moment not continue doctoral studies. I have MS degrees in chemistry and in systems science along with graduate certificates in sustainability and statistics. In addition to the 11 required classes in statistics for the graduate certificate in statistics, I've taken classes in machine learning and in reconstructability analysis, an information-theoretic form of discrete multivariate modeling and data mining. I am bilingual and bicultural. I have worked as an environmental regulatory compliance specialist, a science teacher, and most recently as a project manager with a small team of programmers performing upgrades to the landscape ecological model, LANDIS-II. Currently I am doing a small bit of consulting work as the owner of Northwest Data Science, LLC, and have performed financial risk assessments using mixed, nonparametric-parametric methodologies (multivariate Monte Carlo simulation).

I've done a fair amount of basic data science work and have written quite a few reports as well as two NSF proposals. As a consultant (*pro bono*) for the United Way of the Columbia-Willamette I wrote the narrative (i.e., the rationale and the technical basis) for the Machine Learning Pilot Project (MLPP). The MLPP proposed to use supervised learning algorithms (Naive Bayes classifiers) to assist United Way's partner organizations in identifying combinations of services that could lead to successful outcomes in the "Breaking the Cycle of Childhood Poverty" campaign. I also performed the first MLPP investigation into the use of a Naive Bayes classifier on a typical social services dataset. For this task I used the Python-based, "MultinomialNB" available in the "scikit-learn" library. A colleague and I used a hydrological model (SWAT) to investigate how potential climate, land use, and land cover changes could impact freshwater ecosystem services in the Clackamas River Basin, Oregon. The process-based, spatially explicit SWAT model required weather data projections. I used Python and OPeNDAP (Open-source Project for a Network Data Access Protocol) to extract weather data projections (as NetCDF or .nc files) from the University of Idaho THREDDS Data Server. Additionally, I have queried a large dataset from the USDA National Agricultural Statistics Service (16380077 records on 39 fields for the years 1866-2018; 5.5 Gb) for

[7] Joshua M. Epstein and 2050 Project. *Growing artificial societies: social science from the bottom up.* Complex adaptive systems. Washington, D.C: Brookings Institution Press, 1996.

[8] Stephen Wolfram. *A new kind of science.* Champaign, IL: Wolfram Media, 2002. 1197 pp.

[9] John Harte. *Maximum entropy and ecology: a theory of abundance, distribution, and energetics.* Oxford series in ecology and evolution. Oxford ; New York: Oxford University Press, 2011.

[10] Judea Pearl. *Causality: models, reasoning, and inference.* Cambridge, U.K. ; New York: Cambridge University Press, 2000.

the geospatial analysis of crops in the conterminous US. I used R for the dataset queries, the geospatial analysis, and the subsequent map making.

I am familiar with scripting. On a proficiency level scale of 1 (beginner) to 5 (expert), I am a level 4 (advanced) user of R. Additionally, I am an open-source advocate and have run Linux on my home computers for years. As you may know, this means that I've developed decent, Linux command line capabilities. I prefer data in ASCII text files (.csv format) and use "Awk" and "sed" for data cleaning and pre-processing operations. I have some familiarity with SAS and SQL.

Ultimately, though, I've only just explored the tip of the data science iceberg. There is a lot more to learn (Tableau? Hadoop?) and your enterprise seems to offer a very generous opportunity to due so. It seems to me that we might forge a nice mutualism of our own. Truth be told, I'm not much of a business person and would rather spend my time tackling those ever-present "wicked" problems.

Please do let me know if you'd like to chat or if you'd like to see portfolio items, work samples, scripts, etc. I live just up the road. Thanks so much.

En paz,

BMarron

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